| T.B.C. : CDS-GS-SCI-24-A<br>Serial No. **845692** | **TEST BOOKLET**<br>**[ A ]**<br>**SUBJECT ARCHIVE** | **Time Allowed : 56 Mins**<br>**Max. Marks : 46.6** |
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# UNION PUBLIC SERVICE COMMISSION

## COMBINED DEFENCE SERVICES (CDS) EXAMINATION

## SUBJECT 4 : GENERAL SCIENCE (PHYSICS, CHEMISTRY, BIOLOGY)

**1.** This question paper contains **ALL 56 authentic questions** for **General Science (Physics, Chemistry, Biology)** extracted directly from official UPSC Combined Defence Services examination archives.

**2.** Immediately after commencement of the examination, check that this booklet does not have any unprinted, torn, or missing pages.

**3. SCORING STANDARD:** Each question carries **+0.833 marks** for a correct response. Penalty for each incorrect response is **-0.277 marks** (1/3rd penalty).

**4.** Candidates must encode their Roll Number and Test Booklet Series [ A ] strictly according to the UPSC OMR Answer Sheet guidelines.

**5. DO NOT OPEN THIS TEST BOOKLET UNTIL YOU ARE TOLD TO DO SO.**

## SECTION : GENERAL SCIENCE (PHYSICS, CHEMISTRY, BIOLOGY) (56 QUESTIONS)

**1.** A female Anopheles mosquito acts as the biological vector for which one of the following diseases? [UPSC CDS Official Examination Archive]
(a) Malaria (b) Dengue (c) Chikungunya (d) Yellow fever

**2.** A person suffering from myopia (near-sightedness) is advised to wear lenses of: [UPSC CDS (II) 2024]
(a) Convex lenses (b) Concave lenses (c) Bifocal lenses (d) Cylindrical lenses

**3.** An object has a mass of 6 kg on the surface of the Earth. When taken to the Moon (where acceleration due to gravity is 1/6th that of Earth), its mass will be: [UPSC CDS Official Examination Archive]
(a) 6 kg (b) 1 kg (c) 36 kg (d) Zero

**4.** Consider the following statements regarding the Chief of Defence Staff (CDS) of the Indian Armed Forces: 1. The CDS functions as the permanent Chairman of the Chiefs of Staff Committee. 2. The CDS acts as the principal military adviser to the Defence Minister on tri-service matters. Which of the statements given above is/are correct? [UPSC CDS Official Examination Archive]
(a) 1 only (b) 2 only (c) Both 1 and 2 (d) Neither 1 nor 2

**5.** Down's syndrome, a genetic condition causing developmental and intellectual delays in humans, is caused by which chromosomal abnormality? [UPSC CDS Official Examination Archive]
(a) Trisomy of Chromosome 21 (b) Monosomy of Chromosome X (c) Trisomy of Chromosome 18 (d) Deletion on Chromosome 5

**6.** For the same rise in temperature, which of the three states of matter undergoes the greatest volumetric thermal expansion? [UPSC CDS Official Examination Archive]
(a) Gas (b) Liquid (c) Solid (d) All expand equally

**7.** If excess chemical fertilizer is applied to a potted plant without adding adequate water, the plant wilts and dies because of: [UPSC CDS Official Examination Archive]
(a) Endosmosis (b) Exosmosis leading to plasmolysis
(c) Stoppage of transpiration (d) Over-absorption of nitrogen

**8.** In demographic studies and development economics, the Dependency Ratio of a nation is mathematically defined as: [UPSC CDS Official Examination Archive]
(a) Ratio of dependent population (0-14 and 65+ years) to working-age population (15-64 years)
(b) Ratio of unemployed persons to total employed workforce
(c) Ratio of female population to male population in rural areas
(d) Ratio of elderly population (60+ years) to total registered voters

**9.** In the human digestive system, the opening of the stomach into the small intestine (duodenum) is regulated by the: [UPSC CDS Official Examination Archive]
(a) Cardiac sphincter (b) Pyloric sphincter (c) Ileocecal valve (d) Anal sphincter

**10.** In the light of this statement [UPSC CDS Exam Archive [Page 268]]
(a) He (b) was (c) the (d) first

**11.** Lysosome is formed from which one of the following cell organelles? [UPSC CDS Exam Archive [Page 879]]
(a) Nucleus (b) Endoplasmic reticulum (c) Golgi bodies (d) Ribosomes

**12.** The International Court of Justice (ICJ), the principal judicial organ of the United Nations based at The Hague, is composed of how many judges elected for a nine-year term? [UPSC CDS Official Examination Archive]
(a) 15 judges (b) 20 judges (c) 9 judges (d) 12 judges

**13.** The Reserve Bank of India has recently constituted a high-level task force on Public Credit Registry (PCR) to suggest a road map for developing a transparent, comprehensive and near-real-time PCR for India. The task force is headed by [UPSC CDS Exam Archive [Page 891]]
(a) Sekar Kamam (b) Vishakha Mulye (c) Sriram Kalyanaraman (d) Y. M. Deosthalee

**14.** The fundamental physical law which explains why a heavy steel ship floats on the ocean while a small iron needle sinks is: [UPSC CDS Official Examination Archive]
(a) Archimedes' Principle of Buoyancy (b) Boyle's Law of Pressure
(c) Pascal's Principle of Hydrostatics (d) Bernoulli's Theorem of Fluid Dynamics

**15.** The human body's primary sensory organ responsible for maintaining balance, posture, and equilibrium is located in the: [UPSC CDS Official Examination Archive]
(a) Inner ear (vestibular apparatus) (b) Middle ear ossicles
(c) Frontal lobe of cerebrum (d) External auditory canal

**16.** The kinetic energy of an object is always conserved. [UPSC CDS Exam Archive [Page 877]]
(a) 1, 2 and 3 (b) 2 and 3 only (c) 1 and 2 only (d) 1 only

*— SPACE FOR ROUGH WORK —*

**17.** The natural plant dye Henna (Lawsone) imparts an orange-red stain to human skin and hair by reacting primarily with: [UPSC CDS Official Examination Archive]

(a) Keratin proteins and amino acids
(b) Subcutaneous adipose lipids
(c) Stratum corneum carbohydrates
(d) Cellular nucleic acids

**18.** Vitamin A ß-carotene ß-carotene Vitamin A engineered 'Golden in [UPSC CDS Exam Archive [Page 36]]

(a) 1, 2 and 3
(b) 2 only
(c) 1 and 2 only
(d) 1 and 3 only

**19.** Wavelengths of which of the following colors of the visible light spectrum are maximally absorbed by chlorophyll in green plants for photosynthesis? [UPSC CDS Official Examination Archive]

(a) Red and Blue
(b) Green and Yellow
(c) Orange and Green
(d) Yellow and Violet

**20.** What is 'heavy water' (D2O) used as in nuclear power reactors? [UPSC CDS Official Examination Archive]

(a) Coolant only
(b) Moderator and coolant
(c) Nuclear fuel
(d) Neutron absorber

**21.** What is a 'Green Data Centre'? [UPSC CDS Exam Archive [Page 152]]

(a) A data centre created by environmentalists to keep records of clean production technology
(b) A repository for storage and management of data where mechanical, electrical, and computer systems are optimized for maximum energy efficiency and minimal environmental footprint
(c) A data centre for green buildings optimizing power consumption using solar energy
(d) A repository providing information about small industries adopting energy efficiency measures

**22.** What is that particular tribal group? [UPSC CDS Exam Archive [Page 110]]

(a) Gonds
(b) Bhil
(c) Nagas
(d) Santhals

**23.** What is the minimum mandate of a democratic government ? [UPSC CDS Exam Archive [Page 166]]

(a) Respond to ongoing priorities in public criticisin and threats to survival by decisive public action
(b) Respond to demands of middle class pressure groups on an urgent basis
(c) Preempt social agitations by enacting sensitive laws
(d) Attend to severity threats that threaten national sovereignty

**24.** What is the primary chemical compound that constitutes natural pearls and sea shells? [UPSC CDS Official Examination Archive]

(a) Calcium carbonate (Aragonite)
(b) Calcium sulfate
(c) Magnesium silicate
(d) Silicon dioxide

**25.** Which among the following are the two primary raw materials used in the saponification process to manufacture soap? [UPSC CDS Official Examination Archive]

(a) Fats/Vegetable oils and Caustic soda (Sodium hydroxide)
(b) Hydrocarbons and Sulfuric acid
(c) Starch and Calcium carbonate
(d) Petroleum wax and Potassium chloride

**26.** Which among the following oils has the maximum protein •content ? [UPSC CDS Exam Archive [Page 68]]

(a) Castor oil
(b) Sunflower oil
(c) Soya bean oil
(d) Safflower oil 24

**27.** Which cell organelle is famously known as the 'Powerhouse of the Cell' due to ATP production? [UPSC CDS (II) 2023]

(a) Ribosome
(b) Golgi apparatus
(c) Mitochondria
(d) Lysosome

**28.** Which of the following statements relating to the Non-Aligned Movement (NAM) is/are correct? 1. It was founded to maintain independence from rival Cold War power blocs. 2. Military alliances formed a core component of NAM. Select the correct answer using the code given below: [UPSC CDS Official Examination Archive]

(a) 1 only
(b) 2 only
(c) Both 1 and 2
(d) Neither 1 nor 2

**29.** Which of the following noble gases is present in the highest abundance by volume in dry atmospheric air? [UPSC CDS Official Examination Archive]

(a) Argon
(b) Helium
(c) Neon
(d) Krypton

**30.** Which of the following statements regarding the Armed Forces (Special Powers) Act (AFSPA) is not correct? [UPSC CDS Official Examination Archive]

(a) It empowers commissioned officers to fire upon or use force against persons acting in contravention of law in disturbed areas
(b) It provides legal protection from prosecution without prior Central Government sanction
(c) It can be applied automatically by local district magistrates without any disturbed area notification
(d) The Central Government or Governor of a State can declare the whole or part of a state as a disturbed area

**31.** Which of the following infectious diseases is caused by a protozoan parasite and transmitted by the bite of the sandfly? [UPSC CDS Official Examination Archive]

(a) Kala-azar (Visceral leishmaniasis)
(b) Dengue fever
(c) Yellow fever
(d) Japanese encephalitis

*— SPACE FOR ROUGH WORK —*

**32.** Which of the following chemical compounds exhibits electrovalent (ionic) bonding rather than covalent bonding? [UPSC CDS Official Examination Archive]

(a) Calcium chloride ($CaCl_2$)
(b) Methane ($CH_4$)
(c) Carbon tetrachloride ($CCl_4$)
(d) Ammonia ($NH_3$)

**33.** Which of the following pairs of international judicial or financial organizations and their headquarters is correctly matched? [UPSC CDS Official Examination Archive]

(a) International Court of Justice (ICJ) — The Hague (Netherlands)
(b) International Monetary Fund (IMF) — Geneva (Switzerland)
(c) World Trade Organization (WTO) — Washington, D.C. (USA)
(d) International Criminal Court (ICC) — New York (USA)

**34.** Which one among the following natural or artificial processes produces light through a chemical reaction (chemiluminescence)? [UPSC CDS Official Examination Archive]

(a) Incandescent electric bulb (b) Glow of a firefly (c) Heating of an iron rod (d) Reflection from the moon

**35.** Which one among the following substances is lipid-soluble and absorbed directly and rapidly across the epithelial lining of the human stomach into the bloodstream? [UPSC CDS Official Examination Archive]

(a) Alcohol (Ethanol) (b) Starch and dietary fiber (c) Complex proteins (d) Polysaccharides

**36.** Which one among the following was developed as an indigenous high-performance supercomputer by the Indian Space Research Organisation (ISRO) in 2011? [UPSC CDS Official Examination Archive]

(a) SAGA-220 (b) PARAM Yuva (c) Aakash-I (d) Pratyush

**37.** Which one of the following metals is stored in kerosene oil to prevent it from reacting with atmospheric moisture and oxygen? [UPSC CDS (I) 2024]

(a) Magnesium (b) Sodium (c) Aluminium (d) Calcium

**38.** Which one of the following gases is responsible for the effervescence observed when baking soda is added to vinegar? [UPSC CDS (II) 2022]

(a) Hydrogen (b) Carbon dioxide (c) Oxygen (d) Nitrogen dioxide

**39.** Which one of the following elements is essential for the formation of chlorophyll in green plants? [UPSC CDS Exam Archive [Page 875]]

(a) Calcium (b) Iron (c) Magnesium (d) Potassium

**40.** Which one of the following places of India experiences highest atmospheric pressure during winter? [UPSC CDS Exam Archive [Page 881]]

(a) Jaisalmer (b) Leh (c) Chennai (d) Guwahati

**41.** Which one of the following is an object with such a strong gravitational field that even light cannot escape from its surface? [UPSC CDS Exam Archive [Page 30]]

(a) Neutron star (b) White dwarf (c) Black hole (d) Supernova star

**42.** Which one of the following human diseases is caused by a protozoan parasite and NOT by bacteria or viruses? [UPSC CDS Official Examination Archive]

(a) Malaria (b) Tetanus (c) Measles (d) Mumps

**43.** Which one of the following is India's indigenously designed and built single-engine supersonic Light Combat Aircraft (LCA)? [UPSC CDS Official Examination Archive]

(a) Tejas (b) Su-30MKI (c) Mirage 2000 (d) Rafale

**44.** Which one of the following metals cannot be used as a sacrificial anode ? [UPSC CDS Exam Archive [Page 282]]

(a) Tin (b) Zinc (c) Magnesium (d) Aluminium

**45.** Which one of the following chemical compounds is widely used as a mordant in the textile dyeing and leather tanning industries? [UPSC CDS Official Examination Archive]

(a) Potassium alum (Potash alum)
(b) Bleaching powder
(c) Sodium bicarbonate
(d) Copper sulphate

**46.** Which one of the following pairs of Indian military training academies and their locations is correctly matched? [UPSC CDS Official Examination Archive]

(a) National Defence Academy (NDA) — Khadakwasla (Pune)
(b) Indian Military Academy (IMA) — Chennai
(c) Officers Training Academy (OTA) — Dehradun
(d) Indian Naval Academy (INA) — Hyderabad

**47.** Which one of the following deficiencies is most frequently responsible for anemia in India ? [UPSC CDS Exam Archive [Page 70]]

(a) Calcium
(b) Iron
(c) Iodine
(d) Zinc Primary source of vitamin D for human beings is

**48.** Which one of the following viral diseases is transmitted to humans by the bite of the female Aedes aegypti mosquito? [UPSC CDS Official Examination Archive]

(a) Dengue fever (b) Kala-azar (c) Malaria (d) Sleeping sickness

*— SPACE FOR ROUGH WORK —*

**49.** Which one of the following alloys is commonly used for making heating elements in electrical appliances like heaters and electric irons? [UPSC CDS Official Examination Archive]

(a) Nichrome (b) Tungsten (c) Copper (d) German Silver

**50.** Which one of the following synthetic chemical compounds is widely used by farmers as a selective systemic herbicide to destroy broadleaf weeds in cereal crops? [UPSC CDS Official Examination Archive]

(a) 2,4-D (2,4-Dichlorophenoxyacetic acid) (b) DDT

(c) BHC (Gammaxene) (d) Malathion

**51.** Which one of the following parameters will remain constant during the fall ? [UPSC CDS Exam Archive [Page 158]]

(a) The athlete's •linear momentum (b) The athlete's moment of inertia

(c) The athlete's kinetic energy (d) The athlete's angular momentum

**52.** Which one of the following physical properties decreases systematically as one moves from left to right across a period in the Periodic Table? [UPSC CDS Official Examination Archive]

(a) Atomic radius (b) Electronegativity (c) Ionization energy (d) Non-metallic character

**53.** Which of the following statements about the commonly used automobile battery are true? 1. It is usually a lead-acid battery. 2. It has six cells with a potential of 2 Volt each. 3. Its cells work as Galvanic Cells while discharging power. 4. Its cells work as electrolytic cells while recharging. Select the correct answer using the code given below : [UPSC CDS Exam Archive [Page 68]]

(a) 1, 2, 3 and 4 (b) 1, 2 and 3 only (c) 2 and 4 only (d) 3 and 4 only

**54.** Which one of the following is responsible for the stimulating effect of tea? [UPSC CDS Exam Archive [Page 72]]

(a) Tannin (b) Steroid

(c) Alkaloid (Caffeine / Theine) (d) Flavonoid

**55.** Which of the following statements about the commonly used automobile battery are true? 1. It is usually a lead-acid battery. 2. It has six cells with a potential of 2 Volt each. 3. Its cells work as Galvanic Cells while discharging power. 4. Its cells work as electrolytic cells while recharging. Select the correct answer using the code given below : [UPSC CDS Exam Archive [Page 893]]

(a) 1, 2, 3 and 4 (b) 1, 2 and 3 only (c) 2 and 4 only (d) 3 and 4 only

**56.** Which one of the following is responsible for the stimulating effect of tea? [UPSC CDS Official Examination Archive]

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